



The Plasma Membrane

Dr. Niyan Jassim Mohammad

MBChB, MSc medical physiology

Department of medical physiology and microbiology

College of medicine

Hawler medical university

2025-2026

OBJECTIVES

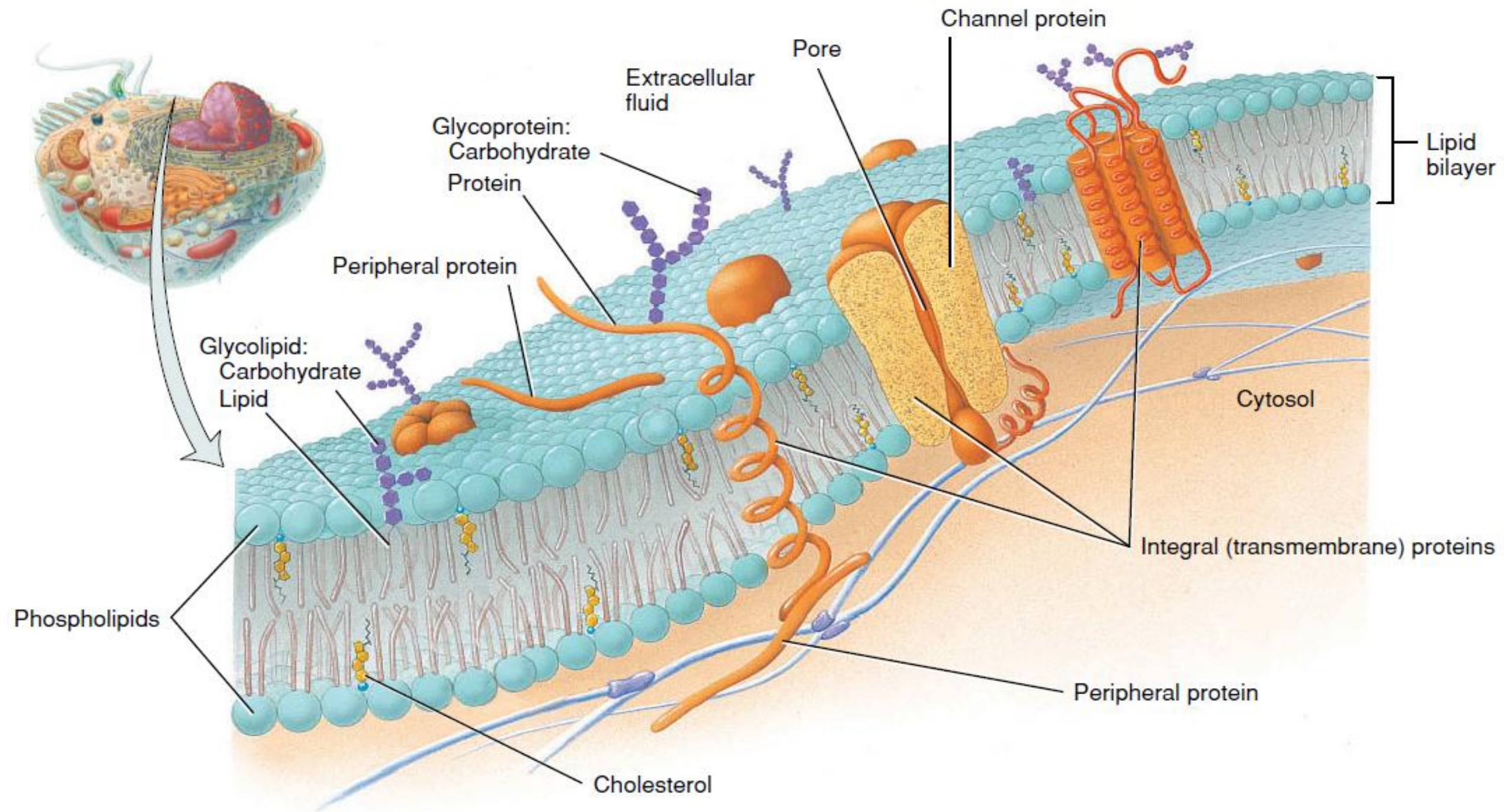
- **Define** the plasma membrane and identify its components.
- **Explain** the concept of selective permeability.
- **Define** the electrochemical gradient and **describe** its components

Introduction to plasma membrane:

- **plasma membrane** is a flexible yet sturdy barrier that surrounds and contains the cytoplasm of a cell.
- It is best described by using a structural model called the **fluid mosaic model**. According to this model, the molecular arrangement of the plasma membrane resembles a continually moving sea of fluid lipids that contains a mosaic of many different proteins.

Introduction to plasma membrane (cont..)

- Some proteins float freely in the lipid, whereas others are anchored at specific locations.
- Membrane lipids allow passage of several types of lipid-soluble molecules but act as a barrier to the entry or exit of charged or polar substances.
- Some of the proteins in the plasma membrane allow movement of polar molecules and ions into and out of the cell.
- Other proteins can act as signal receptors or as molecules that link the plasma membrane to intracellular or extracellular proteins.



FUNCTIONS OF THE PLASMA MEMBRANE

1. Acts as a barrier separating inside and outside of the cell.
2. Controls the flow of substances into and out of the cell.
3. Helps identify the cell to other cells (e.g., immune cells).
4. Participates in intercellular signaling.

Structure of the Plasma Membrane:

- The plasma membrane is **lipid bilayer**, two back-to-back layers made up of three types of lipid molecules—phospholipids, cholesterol, and glycolipids.
- About **75% of the membrane lipids are phospholipids, lipids that contain phosphorus**. Present in smaller amounts are **cholesterol (about 20%), a steroid with an attached OOH (hydroxyl) group, and various glycolipids (about 5%), lipids with attached carbohydrate groups**.
- The bilayer arrangement occurs because the lipids are **amphipathic** molecules, which means that they have both polar and nonpolar parts.

Structure of the Plasma Membrane (cont..)

- In phospholipids, the polar part is the phosphate-containing “head,” which is *hydrophilic*. The nonpolar parts are the two long fatty acid “tails,” which are *hydrophobic* hydrocarbon chains.
- The phospholipid molecules orient themselves in the bilayer with their hydrophilic heads facing outward. In this way, the heads face a watery fluid on either side—cytosol on the inside and extracellular fluid on the outside. The hydrophobic fatty acid tails in each half of the bilayer point toward one another, forming a nonpolar, hydrophobic region in the membrane’s interior.

Structure of the Plasma Membrane (cont..)

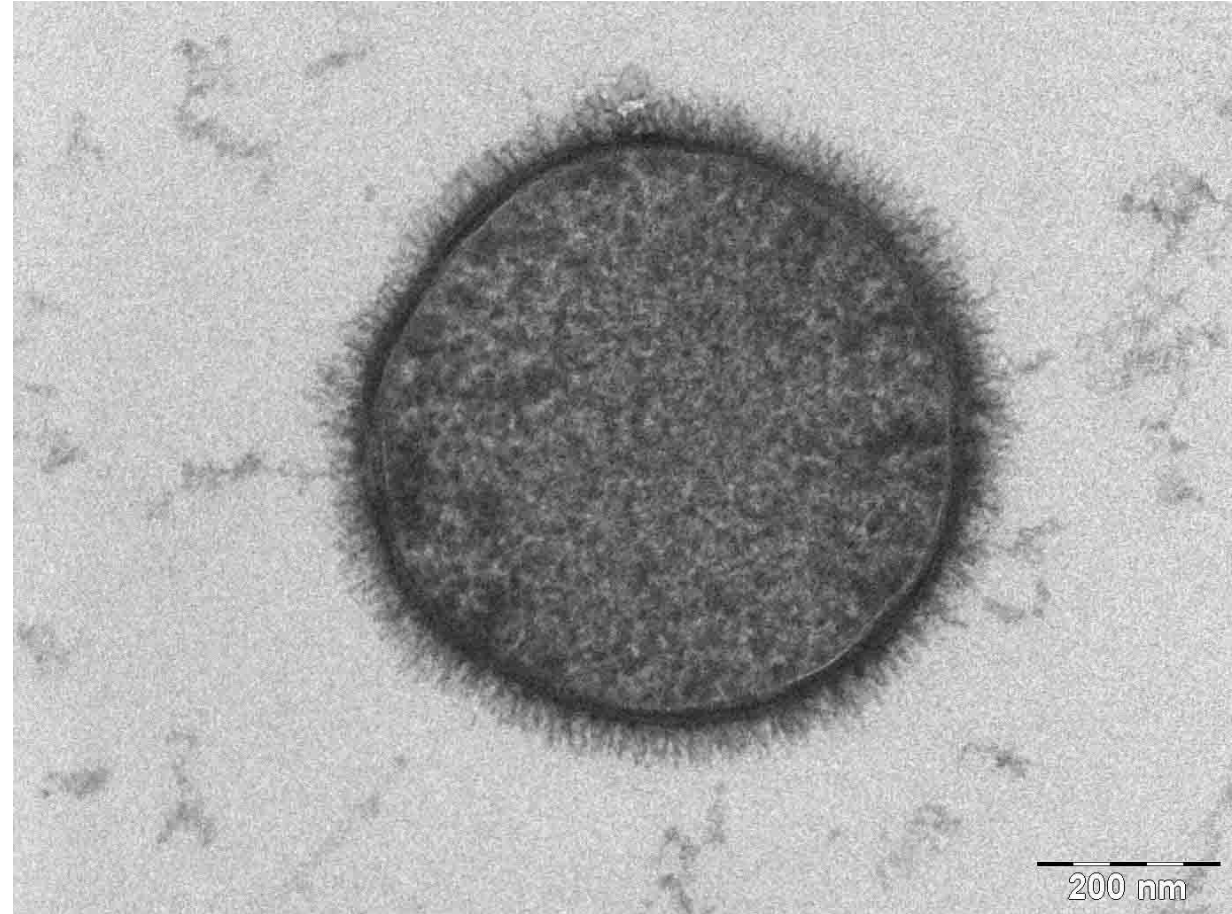
- Cholesterol molecules are weakly amphipathic and are interspersed among the other lipids in both layers of the membrane.
- The tiny —OH group is the only polar region of cholesterol, and it forms hydrogen bonds with the polar heads of phospholipids and glycolipids.
- The stiff steroid rings and hydrocarbon tail of cholesterol are nonpolar; they fit among the fatty acid tails of the phospholipids and glycolipids.
- The carbohydrate groups of glycolipids form a polar “head”; their fatty acid “tails” are nonpolar.
- Glycolipids appear only in the membrane layer that faces the extracellular fluid, which is one reason the two sides of the bilayer are asymmetric, or different.

Arrangement of Membrane Proteins:

- Membrane proteins are classified as integral or peripheral according to whether they are firmly embedded in the membrane.
- **Integral proteins** extend through the lipid bilayer and are firmly embedded in it. Most of them are **transmembrane proteins**, which means that they span the entire lipid bilayer and protrude into both the cytosol and extracellular fluid. A few integral proteins are tightly attached to one side of the bilayer by covalent bonding to fatty acids.
- Like membrane lipids, integral membrane proteins are amphipathic. Their hydrophilic regions protrude into either the watery extracellular fluid or the cytosol, and their hydrophobic regions extend among the fatty acid tails.
- **peripheral proteins** are not as firmly embedded in the membrane. They are attached to the polar heads of membrane lipids or to integral proteins at the inner or outer surface of the membrane

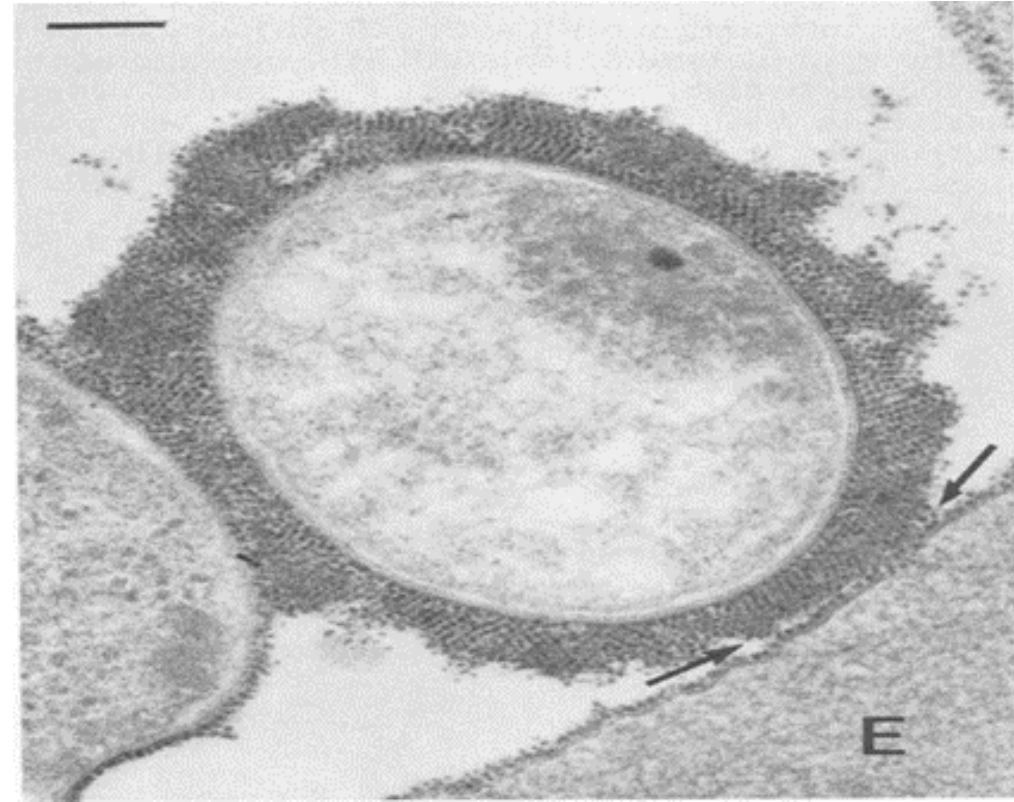
Arrangement of Membrane Proteins (cont..)

- Many integral proteins are **glycoproteins**, proteins with carbohydrate groups attached to the ends that protrude into the extracellular fluid.
- The carbohydrate portions of glycolipids and glycoproteins form an extensive sugary coat called the **glycocalyx**
- The pattern of carbohydrates in the glycocalyx varies from one cell to another.



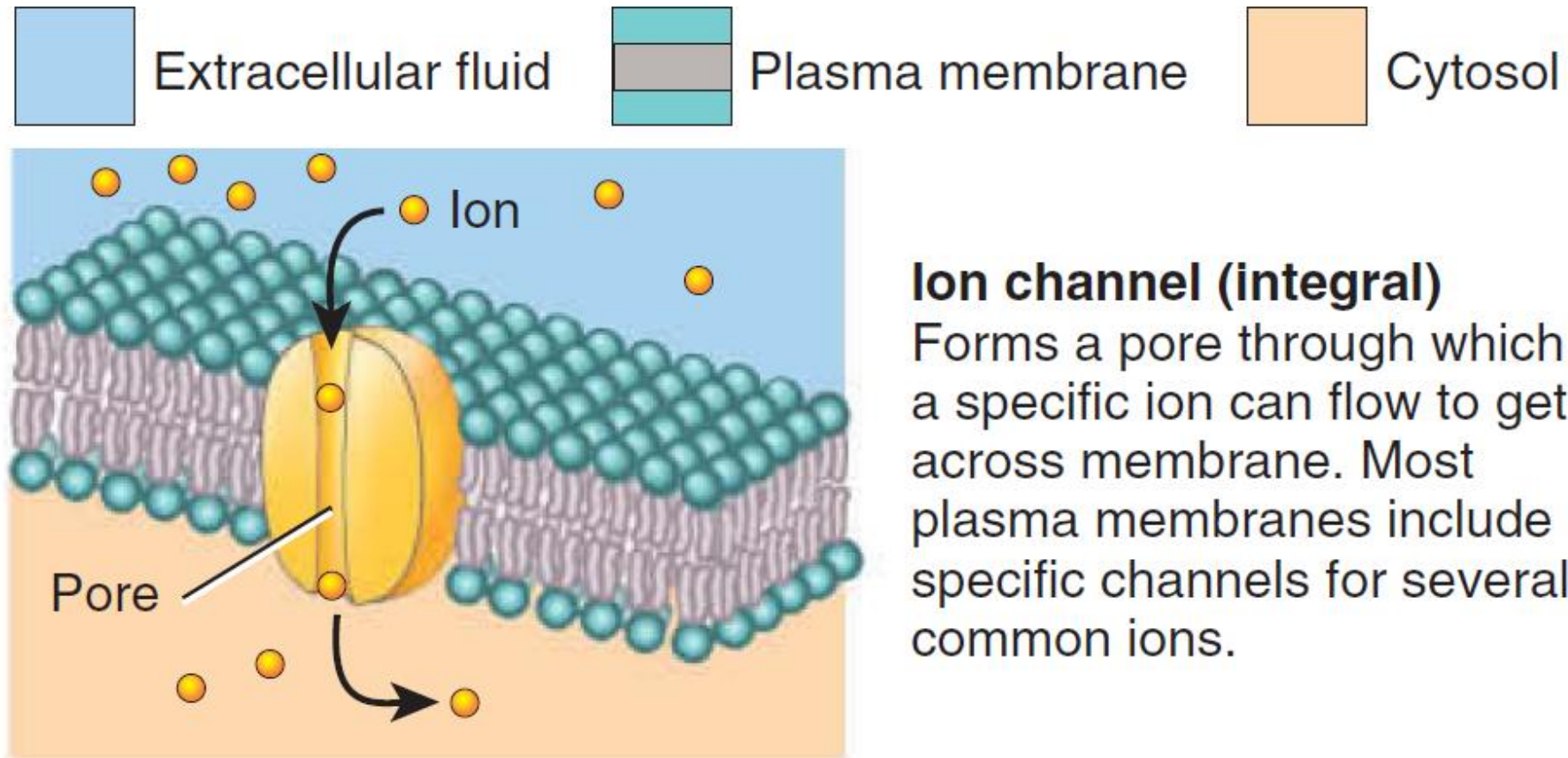
Arrangement of Membrane Proteins (cont..)

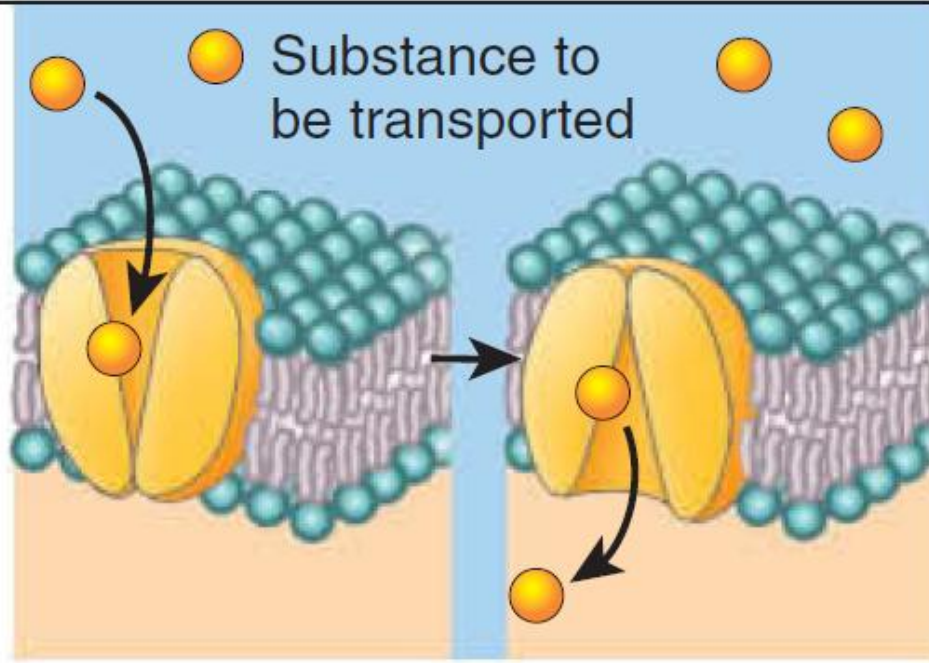
- the glycocalyx acts like a molecular “signature” that enables cells to recognize one another. For example, a white blood cell’s ability to detect a “foreign” glycocalyx is one basis of the immune response that helps us destroy invading organisms.
- the glycocalyx enables cells to adhere to one another in some tissues and protects cells from being digested by enzymes in the extracellular fluid.
- The hydrophilic properties of the glycocalyx attract a film of fluid to the surface of many cells. This makes red blood cells slippery as they flow through narrow blood vessels and protects cells that line the airways and the gastrointestinal tract from drying out.



Functions of Membrane Proteins:

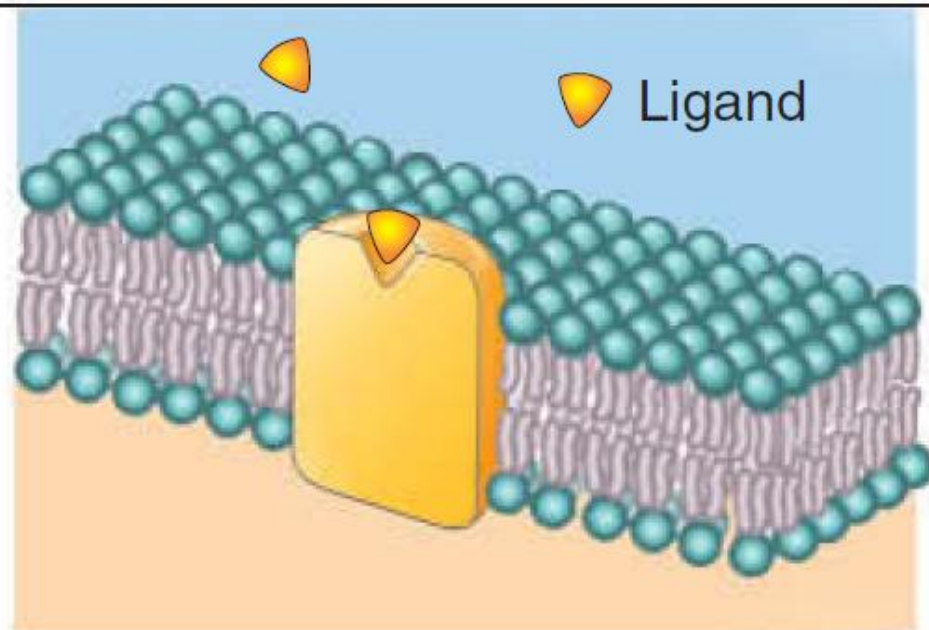
- Generally, the types of lipids in cellular membranes vary only slightly. In contrast, the membranes of different cells and various intracellular organelles have remarkably different assortments of proteins that determine many of the membrane's functions.





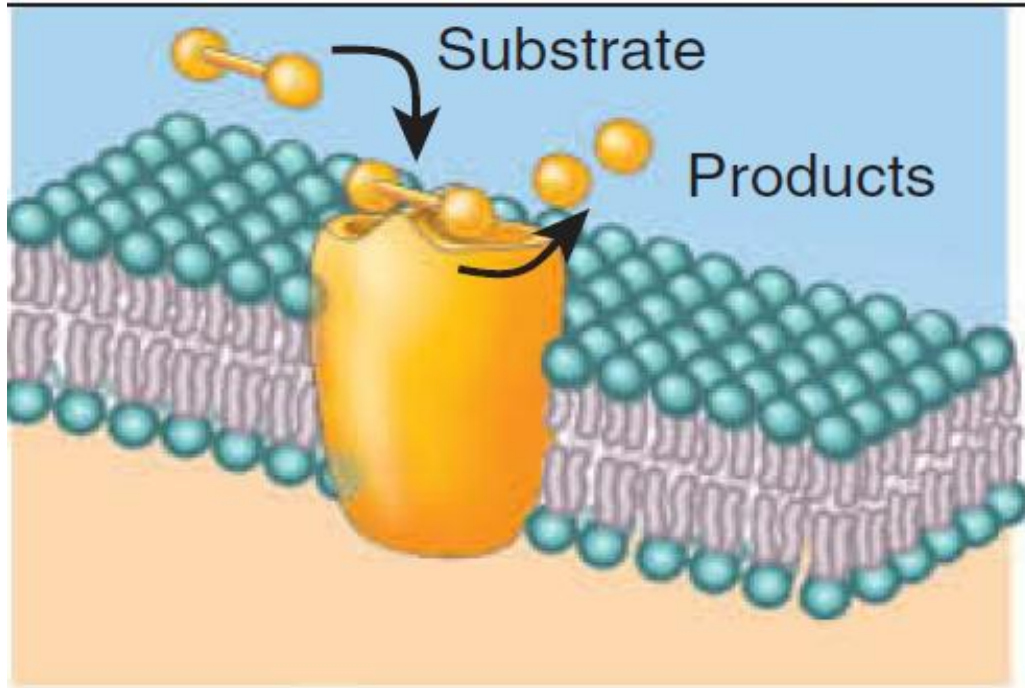
Carrier (integral)

Transports a specific substance across membrane by undergoing a change in shape. For example, amino acids, needed to synthesize new proteins, enter body cells via carriers. Carrier proteins are also known as *transporters*.



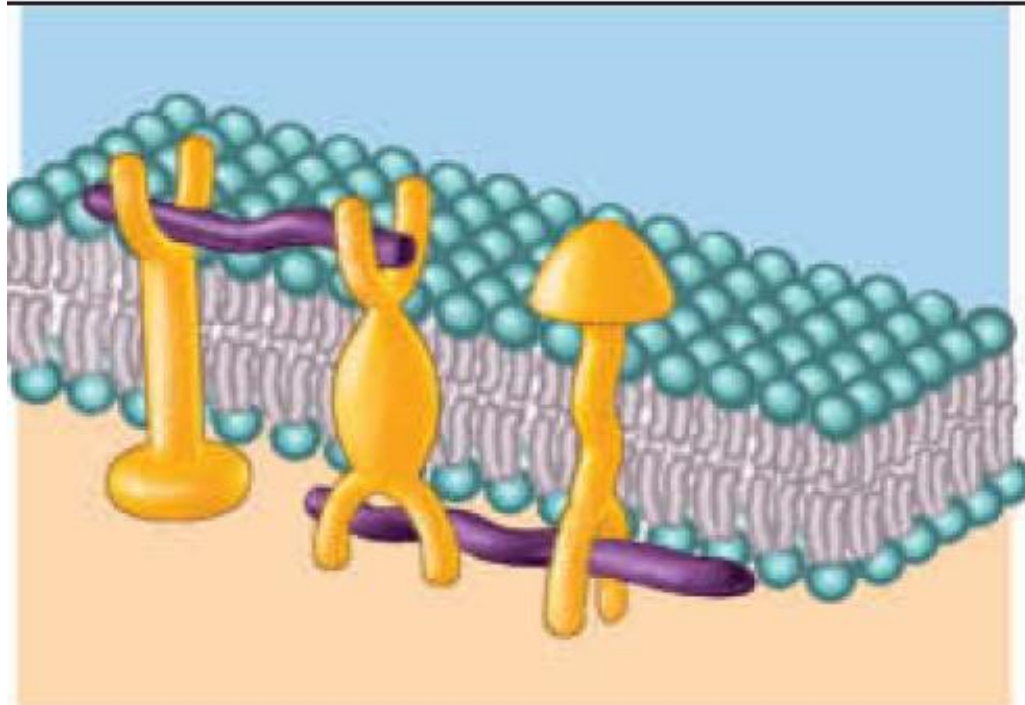
Receptor (integral)

Recognizes specific ligand and alters cell's function in some way. For example, antidiuretic hormone binds to receptors in the kidneys and changes the water permeability of certain plasma membranes.



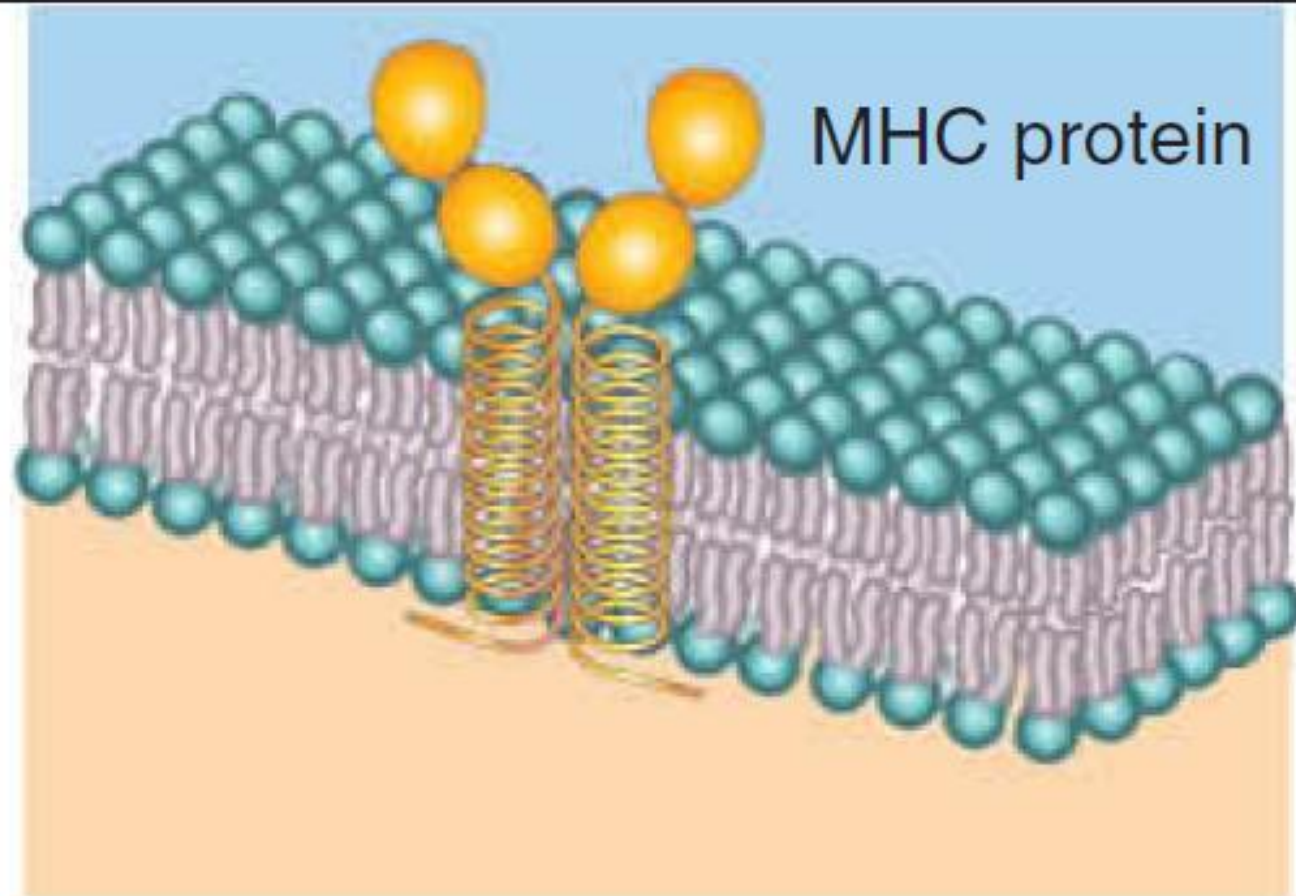
Enzyme (integral and peripheral)

Catalyzes reaction inside or outside cell (depending on which direction the active site faces). For example, lactase protruding from epithelial cells lining your small intestine splits the disaccharide lactose in the milk you drink.



Linker (integral and peripheral)

Anchors filaments inside and outside the plasma membrane, providing structural stability and shape for the cell. May also participate in movement of the cell or link two cells together.



Cell identity marker (glycoprotein)

Distinguishes your cells from anyone else's (unless you are an identical twin). An important class of such markers are the major histocompatibility (MHC) proteins.

Membrane Fluidity

- Membranes are fluid structures; that is, most of the membrane lipids and many of the membrane proteins easily rotate and move sideways in their own half of the bilayer.
- Its depends both on the number of double bonds in the fatty acid tails of the lipids that make up the bilayer, and on the amount of cholesterol present.
- Is an excellent compromise for the cell; a rigid membrane would lack mobility, and a completely fluid membrane would lack the structural organization and mechanical support required by the cell.
- It allows interactions to occur within the plasma membrane, such as the assembly of membrane proteins, it also enables the movement of the membrane components responsible for cellular processes such as cell movement, growth, division, and secretion, and the formation of cellular junctions.
- Fluidity allows the lipid bilayer to self-seal if torn or punctured. When a needle is pushed through a plasma membrane and pulled out, the puncture site seals spontaneously, and the cell does not burst. This property of the lipid bilayer allows a procedure called intracytoplasmic sperm injection to help infertile couples conceive a child; scientists can fertilize an oocyte by injecting a sperm cell through a tiny syringe.

Membrane Permeability:

- The term *permeable* means that a structure permits the passage of substances through it.
- Plasma membranes permit some substances to pass more readily than others. This property of membranes is termed **selective permeability**.
- The lipid bilayer portion of the plasma membrane is highly permeable to nonpolar molecules such as oxygen (O₂), carbon dioxide (CO₂), and steroids; moderately permeable to small, uncharged polar molecules, such as water and urea (a waste product from the breakdown of amino acids); and impermeable to ions and large, uncharged polar molecules, such as glucose.
- The permeability characteristics of the plasma membrane are due to the fact that the lipid bilayer has a nonpolar, hydrophobic interior. So, the more hydrophobic or lipid-soluble a substance is, the greater the membrane's permeability to that substance.
- Thus, the hydrophobic interior of the plasma membrane allows nonpolar molecules to rapidly pass through, but prevents passage of ions and large, uncharged polar molecules.

Membrane Permeability (cont..)

- The permeability of the lipid bilayer to water and urea is an unexpected property given that they are polar molecules.
- These two molecules are thought to pass through the lipid bilayer in the following way. As the fatty acid tails of membrane phospholipids and glycolipids randomly move about, small gaps briefly appear in the hydrophobic environment of the membrane's interior. Because water and urea are small polar molecules that have no overall charge, they can move from one gap to another until they have crossed the membrane.
- Transmembrane proteins that act as channels and carriers increase the plasma membrane's permeability to a variety of ions and uncharged polar molecules that, unlike water and urea molecules, cannot cross the lipid bilayer unassisted.
- Channels and carriers are very selective. Each one helps a specific molecule or ion to cross the membrane. Macromolecules, such as proteins, are so large that they are unable to pass across the plasma membrane except by endocytosis and exocytosis.

Gradients across the Plasma Membrane:

- The selective permeability of the plasma membrane allows a living cell to maintain different concentrations of certain substances on either side of the plasma membrane.
- A **concentration gradient** is a difference in the concentration of a chemical from one place to another, such as from the inside to the outside of the plasma membrane.
- Many ions and molecules are more concentrated in either the cytosol or the extracellular fluid. For instance, oxygen molecules and sodium ions (Na) are more concentrated in the extracellular fluid than in the cytosol; the opposite is true of carbon dioxide molecules and potassium ions (K).
- The plasma membrane also creates a difference in the distribution of positively and negatively charged ions between the two sides of the plasma membrane. Typically, the inner surface of the plasma membrane is more negatively charged and the outer surface is more positively charged.
- A difference in electrical charges between two regions constitutes an **electrical gradient**. Because it occurs across the plasma membrane, this charge difference is termed the **membrane potential**.

Gradients across the Plasma Membrane (cont..)

- The concentration gradient and electrical gradient are important because they help move substances across the plasma membrane.
- In case a substance will move across a plasma membrane *down its concentration gradient*. That is to say, a substance will move “downhill,” from where it is more concentrated to where it is less concentrated, to reach equilibrium.
- Similarly, a positively charged substance will tend to move toward a negatively charged area, and a negatively charged substance will tend to move toward a positively charged area.
- The combined influence of the concentration gradient and the electrical gradient on movement of a particular ion is referred to as its **electrochemical gradient**.



THANK YOU